

Subject Description Form

Subject Code	AP5009
Subject Title	Emerging Memory Technologies
Credit Value	3
Level	5
Pre-requisite/ Co-requisite/ Exclusion	NIL
Objectives	The objective of this subject is to introduce concepts of the memory sub-system from the device cell structures to the array and architecture design with emphasis on the industry trend and cutting-edge technologies. The concept of the memory hierarchy is used as an outline throughout the whole course. The in-memory computing and in-sensor computing for artificial intelligence will be also covered.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: (a) Understand the working principles of conventional charge-based memory technology (b) Understand the working principles of emerging non-charge-based memory technology (c) Understand the fundamentals of in-memory computing and in-sensor computing (d) Understand the working principles of the cross-point memory array for artificial neural network (e) Experimentally carry out the micro-fabrication and characterization of emerging memory using the facilities inside the cleanroom
Subject Synopsis/ Indicative Syllabus	Charge-based memory technology: Static Random Access Memory, Dynamic Random Access Memory, Flash Memory, etc. Non-charge-based memory technology: Resistive switching memory, phase-change memory, magnetic memory, etc. In-memory computing and in-sensor computing: The principles of multiplication and accumulation. Cross-point memory array for the artificial neural network: Study the working principles and design of cross-point memory array for artificial neural network applications.
Teaching/Learning Methodology	Lecture: Basic theory and knowledge will be systematically introduced in lectures. Classwork and assignments related to the content of lectures will be used to enhance students learning.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)				
			a	b	c	d	e
	(1) Continuous assessments	50	✓	✓	✓	✓	✓
	(2) Project	50	✓	✓	✓	✓	✓
	Total	100					
	Continuous assessments will strengthen the students’ basic knowledge and analytical skills to solve the problems related to this subject. The course is a cutting-edge technology-oriented course; thus the students are expected to actively read the related literature as part of the learning process. A project will be assigned to a group of students. Students are expected to use their knowledge gained in the class/lab to perform the design and micro-fabrication of memory devices. Students should have gained the basic knowledge of a) charge-based memory technology and b) non-charge-based memory technology; c) have a thorough understanding of in-memory and in-sensor computing; d) be able to understand the working principle of the memory array for artificial neural networks, and e) be able to carry out the micro-fabrication in our cleanroom.						
Student Study Effort Expected	Class contact:						
	• Lecture/Seminar		21 h				
	Other student study efforts:						
	• Laboratory		18 h				
	• Self-study		81h				
	Total student study effort:		120 h				
Reading List and References	1. Koichiro Ishibashi, Kenichi Osada, Low Power and Reliable SRAM Memory Cell and Array Design, Springer, 2011. 2. Brent Keeth, R. Jacob Baker, Brian Johnson, and Feng Lin, DRAM Circuit Design: Fundamental and High-Speed Topics, 2nd Ed., John Wiley & Sons, 2008 3. Seiichi Aritome, NAND FLASH Memory Technologies, Wiley-IEEE Press, 2016. 4. Shimeng Yu, Resistive Random Access Memory (RRAM), Morgan & Claypool, 2016.						